SHEEHAN GANGULY

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University of Washington - Paul Allen ***September 2023 – June 2027***
Bachelor's, Computer Engineering | GPA: 3.75 ***Intended Masters in ECE - Sep 2027 - June 2028***
Relevant Coursework: ASIC Tapeout Capstone, VLSI, Digital IC Design, FPGA Design, Advanced Semiconductors, Computer Architecture, Semiconductor Devices, Embedded Systems
Languages: SystemVerilog, C/C++, Java, Python, UVM, RISC-V Assembly, Linux CLI
RTL & Digital Design: Synthesizable SystemVerilog, FSMs, Pipelined Architectures, Datapath/Control Design, Clock-Aware Design
ASIC & Verification: Cadence Genus, Innovus, Tempus, Virtuoso; Synopsys VCS/Verdi; UVM, STA, SDC, DRC/LVS
FPGA & Embedded: Xilinx Vivado, Intel Quartus/ModelSim, QuestaSim, Lattice iCEcube2, FABulous, FreeRTOS, UART, LoRaWAN, Intel DE1-SoC

EXPERIENCE:

Lawrence Berkeley National Lab - Mixed-Signal IC Design Intern ***May 2026 - Present***
- Architected the integration of an Ibex RISC-V processor into an AI accelerator SoC, defining connectivity to on-chip memory, peripherals, and reconfigurable compute hardware.
- Integrated SRAM, UART, and eFPGA peripherals through the TL-UL interconnect, developing the memory-mapped architecture and UART path for host-side programming, control, and data transfer.
- Developed an eFPGA compute architecture using FABulous, configuring the reconfigurable fabric and generating synthesizable RTL for SoC integration.
- Verified RISC-V subsystem, peripheral integration, and eFPGA configuration in QuestaSim, validating TL-UL transactions, memory accesses, peripheral communication, and system-level behavior.

Undergraduate Researcher at PNCEL Lab - Hardware-Software Co-Design for AI Systems ***December 2025 - Present***
- Developing DORA, a CGRA-generation framework for exploring specialization and reconfigurability tradeoffs in spatial accelerator architectures.
- Managed design and verification for a full-system TSMC 180nm ASIC tapeout integrating a hybrid CGRA/systolic architecture for AI inference, from RTL through physical design and tapeout.

Hardware Intern at Lawrence Berkeley National Lab ***June 2025 - August 2025***
- Developed low-power embedded firmware and a LoRaWAN sensor platform for remote environmental monitoring deployments.
- Reduced TI mmWave radar power consumption with an Arduino-controlled relay system and implemented FFT-based processing for radar data analysis.

Hardware/Software Intern at S5 Solutions Seattle, WA ***June 2024 – September 2024***
- Built a Node.js/Express backend that processed PCBA Gerber files to identify probable test features and generate probe locations and sizes.
- Implemented FPGA logic with flip-flops and LUTs in VHDL on a Lattice device using iCEcube2, including UART communication with a host computer.

SELECTED HARDWARE PROJECTS:

Dual-Mode INT8 CGRA / INT4 Systolic Accelerator - RTL to Tapeout, TSMC 180nm
- Led physical implementation and contributed external-interface RTL and verification for a 4×4 reconfigurable PE fabric.
- Implemented dual 8-bit data networks, shared multiplier hardware, packed INT4 DOT4 execution, and local 16-bit accumulation within the reconfigurable PE fabric.
- Developed end-to-end Synopsys VCS tests that configured the fabric through BSG-Link, streamed directed and randomized GEMM inputs, and checked returned results against a software golden model.
- Executed synthesis, place-and-route, extraction, and MMMC STA through Hammer using Cadence Genus, Innovus, and Tempus; closed timing at 50 MHz and passed post-synthesis and post-PAR verification.

CMOS Register File Layout – 16×13-bit Register File (Cadence Virtuoso)
- Designed a full-custom 16×13-bit CMOS register file in Cadence Virtuoso, achieving 2 GHz operation.
- Verified schematic and extracted-layout functionality using HSPICE transient simulations to validate register read/write timing.
- Performed DRC and LVS verification, ensuring layout-schematic equivalence and compliance with process design rules.
- Engineered a bit-sliced, row-organized physical layout utilizing edge-triggered D flip-flops, routing wordlines horizontally and clock/data signals vertically to minimize skew and congestion.

ARM-Based 5-Stage Pipelined CPU
- Designed and verified a 5-stage pipelined processor implementing an ARM-based ISA, achieving a 2.7× speedup over a single-cycle baseline across benchmarks including Quicksort and Fibonacci.
- Implemented forwarding, hazard detection, and branch prediction, reducing benchmark CPI by 70% relative to the unoptimized pipeline
- Developed SystemVerilog RTL for the register file, ALU, pipeline datapath/control, and memory hierarchy including cache and data memory.